Finite-h MDP Overview

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User

formulation of finite h MDP

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A finite-h Markov Decision Process (MDP) is a mathematical framework used to model decision-making in stochastic environments where outcomes are partly random and partly under the control of a decision-maker. Here are the key components of a finite-h MDP:

Components of a Finite-h MDP

1. **State Space (S)**:
2.

A finite set of states (S = {s_1, s_2, \ldots, s_n}) representing all possible states of the environment.

3.

Action Space (A):

4.

A finite set of actions (A = {a_1, a_2, \ldots, a_m}) available to the decision-maker.

5.

Transition Probability:

6. A transition function (P(s' | s, a)) that defines the probability of transitioning to state (s') given the current state (s) and action (a). This can be represented as a probability distribution over the next state.

[P: S \times A \times S \rightarrow [0, 1]]

1. **Reward Function (R)**:
2.

A reward function (R(s, a, s')) that gives the immediate reward received after taking action (a) in state (s) and transitioning to state (s').

3.

Horizon (h):

4.

A finite time horizon (h), which means that decisions are made over (h) time steps. After (h) steps, the process terminates.

5.

Policy ((\pi)):

6. A policy (\pi) defines a mapping from states to actions. It can be deterministic ((\pi: S \rightarrow A)) or stochastic ((\pi: S \times A \rightarrow [0, 1]) indicating the probability of taking action (a) in state (s)).

Objective

The objective in a finite-h MDP is typically to find a policy that maximizes the expected total discounted reward or reward over a finite time horizon, which can be formulated as:

[V^\pi(s) = \mathbb{E}\left[\sum_{t=0}^{h-1} R(s_t, a_t, s_{t+1}) \mid s_0 = s, \pi\right]]

where: - (s_t) is the state at time (t), - (a_t = \pi(s_t)) is the action chosen according to policy (\pi), - (R(s_t, a_t, s_{t+1})) is the reward received after transitioning to (s_{t+1}).

Solving Finite-h MDPs

Dynamic Programming: One common approach to solve finite-h MDPs is through backward induction. You can compute the value function for each state at time (h) and then backtrack to find optimal actions.

Algorithms: Several algorithms are used to find optimal policies, including Value Iteration, Policy Iteration, and Linear Programming methods.

Conclusion

The formulation of a finite-h MDP allows for the systematic analysis of decision-making processes in uncertain environments with a clear structure. By defining states, actions, transitions, rewards, and a finite horizon, we can apply various computational methods to derive optimal policies.